



Kontraksi Otot

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Learning Objective

Definisi dan Fungsi Otot

Terminologi

Jenis-Jenis Otot

Mekanisme

Korelasi Klinis

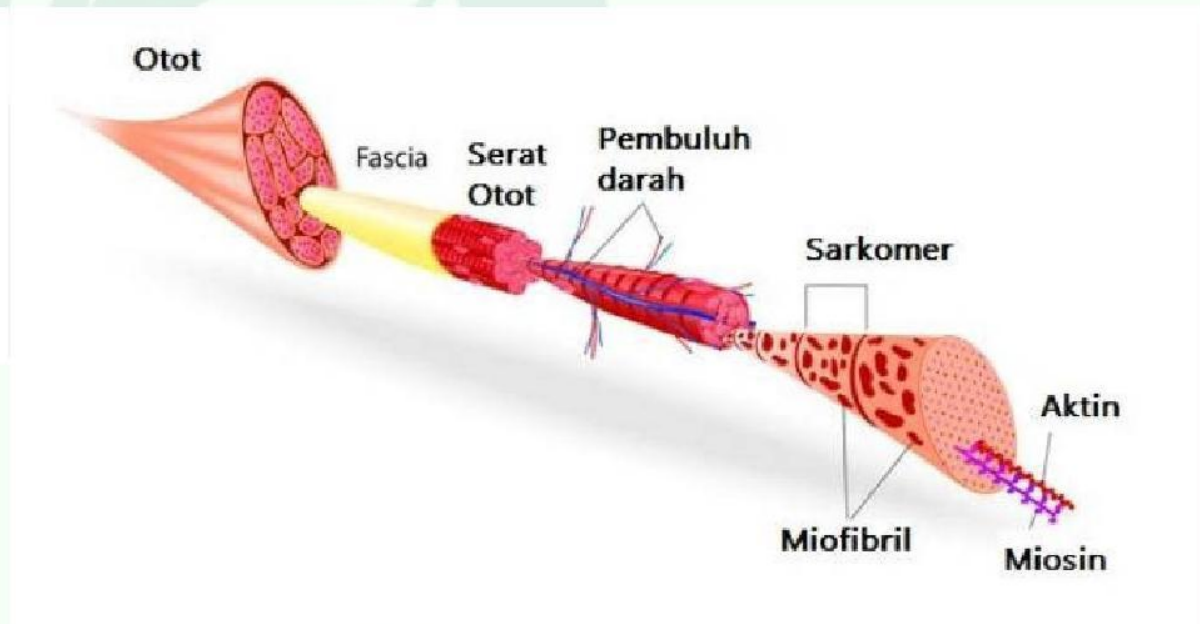
Definisi dan Fungsi Otot



Definisi dan Fungsi Otot

Otot Lurik:

Jaringan berisi beberapa bundel fibrosa (myofiber) yang berkemampuan untuk berkontraksi, menghasilkan gerakan tubuh, dan mempertahankan posisi bagian tubuh.

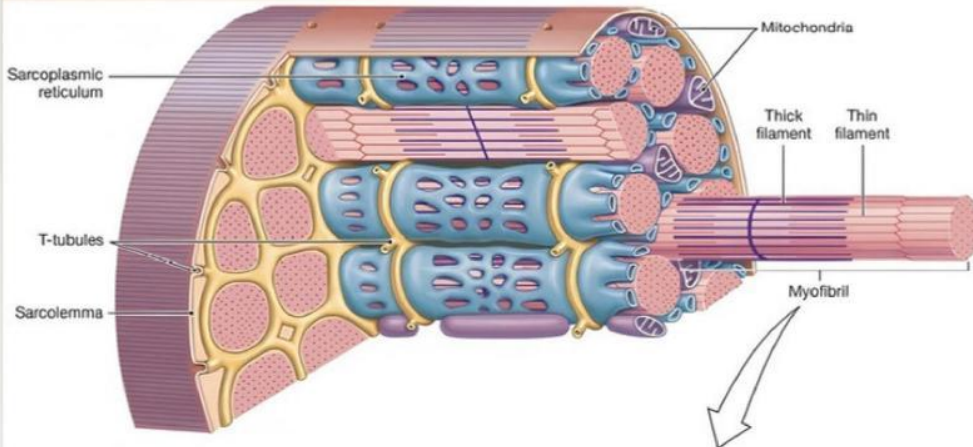


Terminologi



Ultrastructure of Muscle Fiber and Myofibril

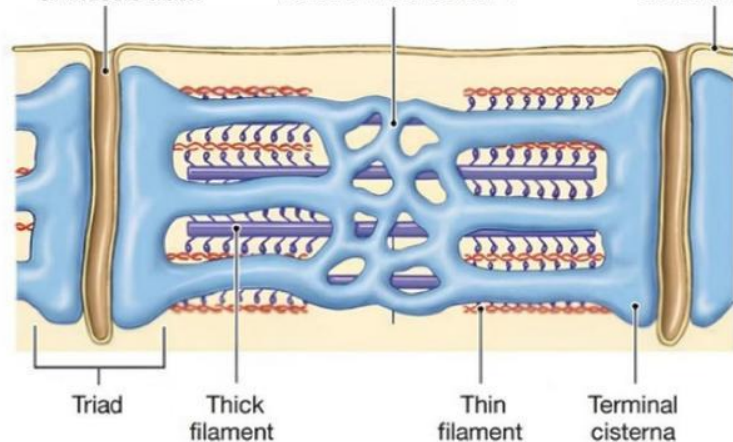
(b) Structure of a Skeletal Muscle Fiber



T-tubule brings action potentials into interior of muscle fiber.

Sarcoplasmic reticulum stores Ca^{2+} .

Sarcolemma



Triad

Thick filament

Thin filament

Terminal cisterna

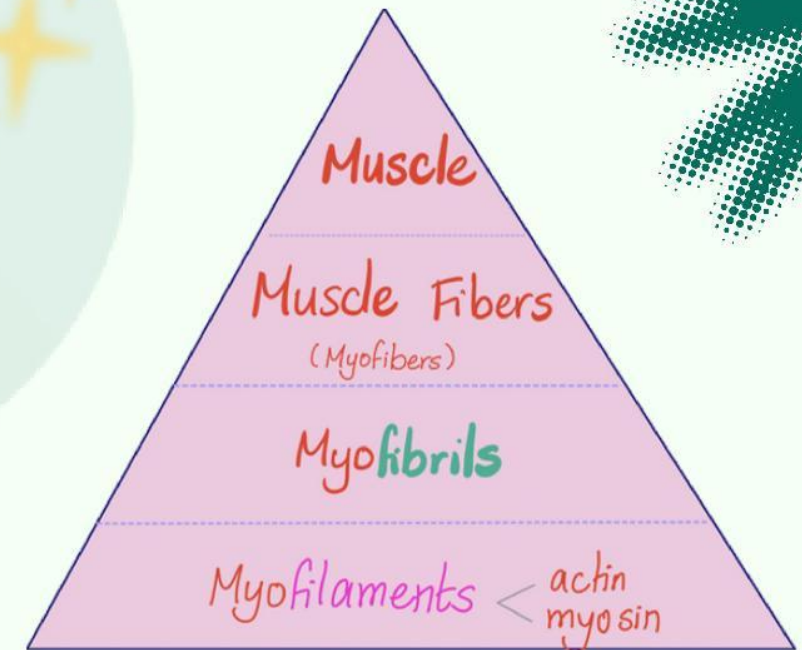
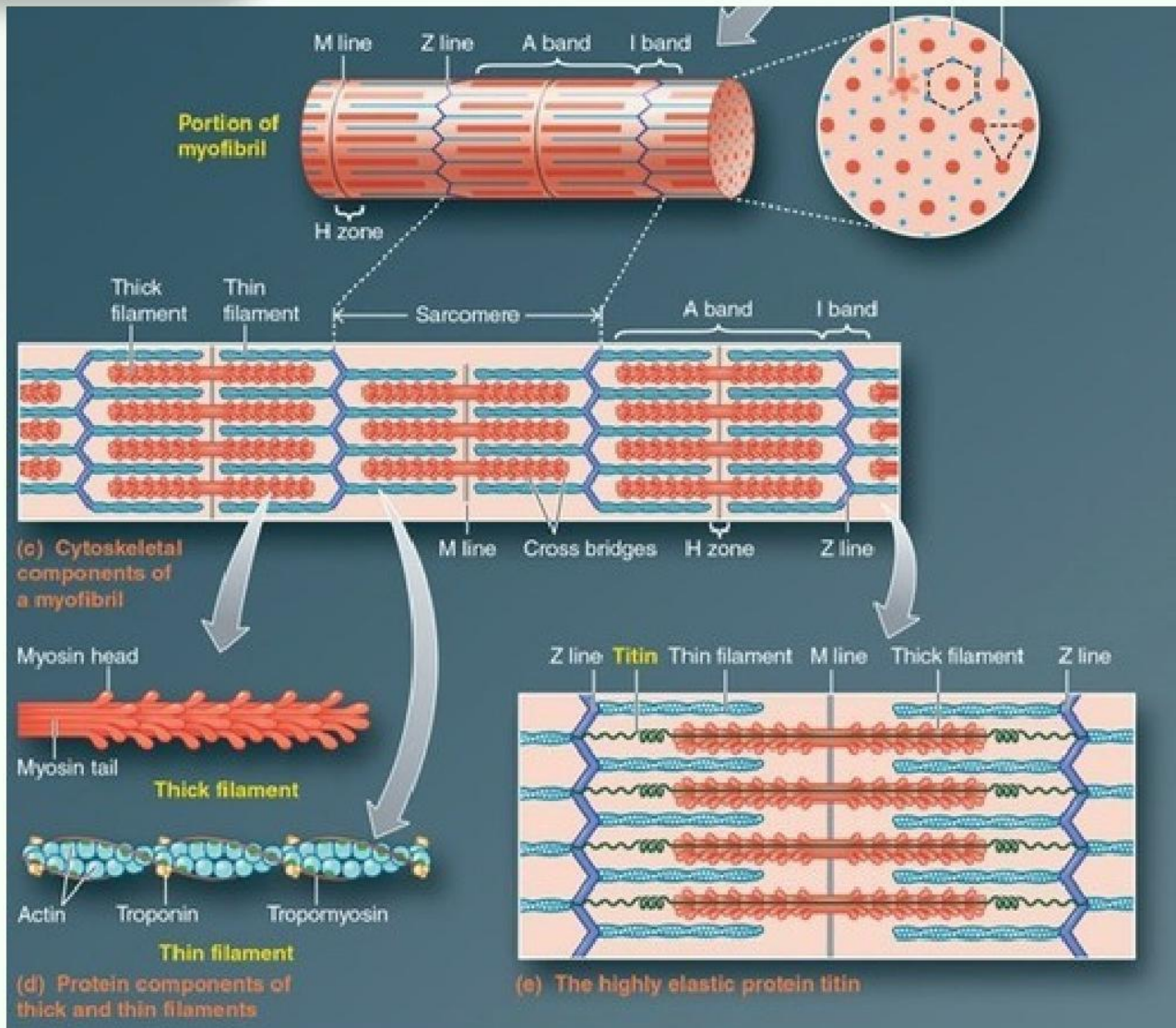
Terminologi

- Sarkolema: membran sel otot
- Sarkoplasma: sitoplasma sel otot
- Retikulum sarkoplasma: Retikulum Endoplasma di sel otot
- Tubulus T (Transversus): Invaginasi /pelipatan ke dalam dari sarkolema yang membentuk seperti kanal-kanal
- Terminal Sisterna: Bagian berbentuk saccus yang merupakan tempat melekatnya retikulum sarkoplasma dengan Tubulus T
- Triad: 1 Tubulus T + 2 Terminal Sisterna

Terminologi

Lapisan Otot Lurik

Sel otot = miosit/
miofiber/serabut otot



Terminologi

Lapisan Otot Lurik

Dari luar ke dalam: Epimisium → Perimisium
→ Endomisium → Sarkolemma

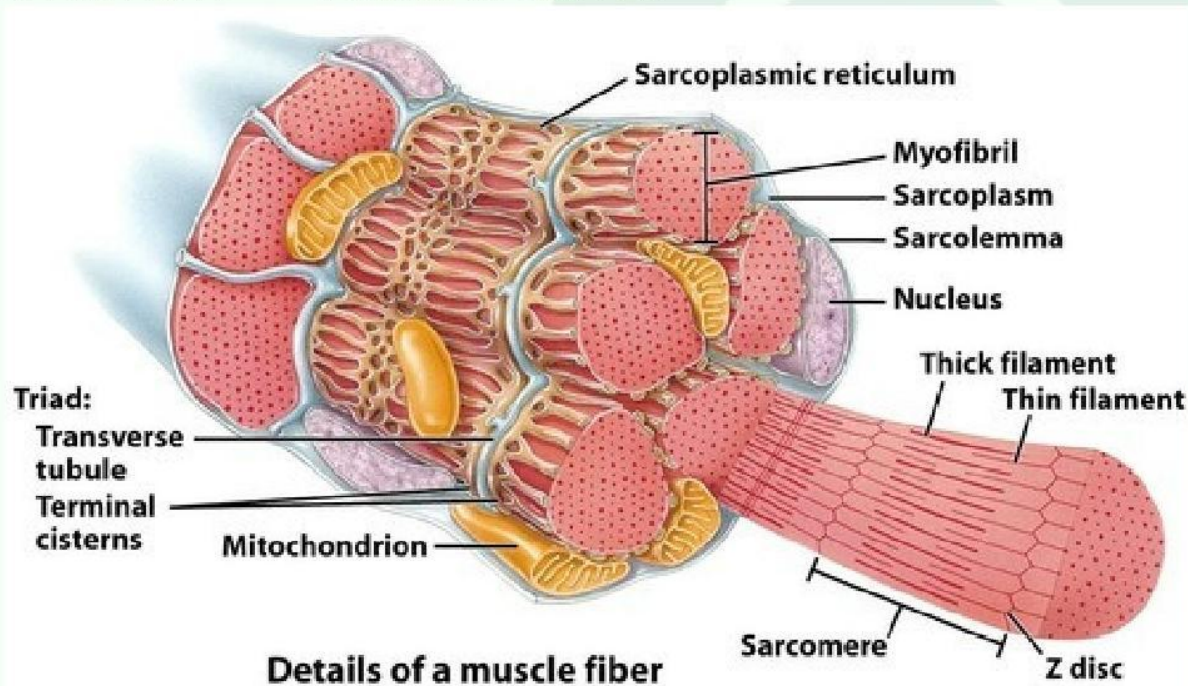
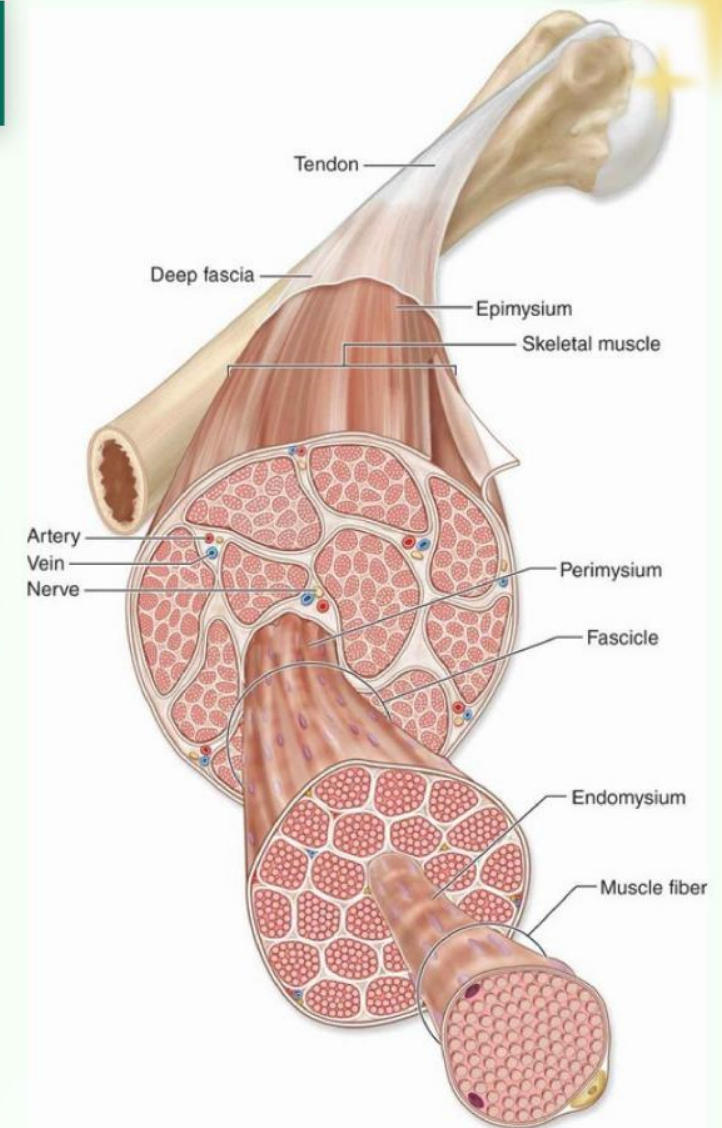


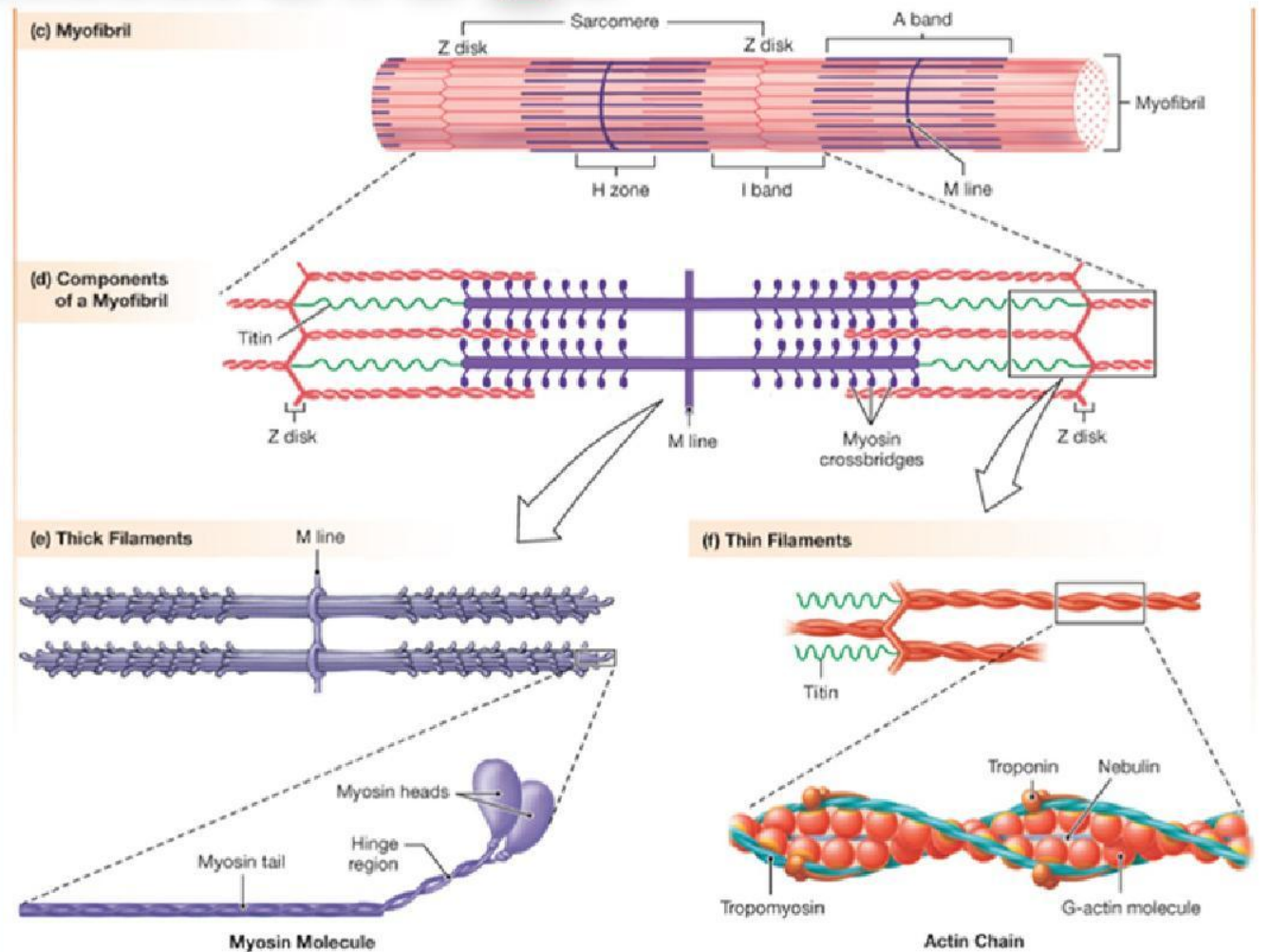
Figure 10-2x. Principles of Anatomy and Physiology, 11/e
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Terminologi

Sarkomer

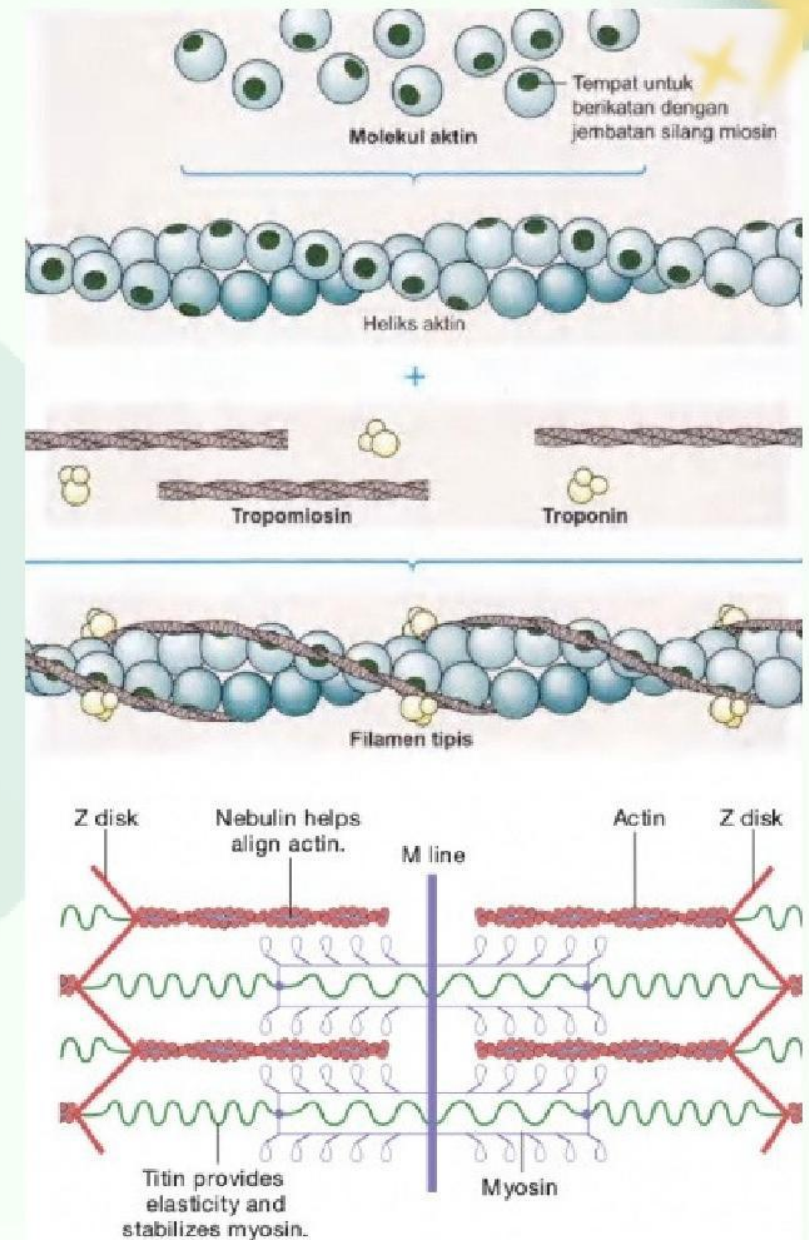
→ Unit fungsional terkecil dari otot lurik yang dibatasi oleh dua garis Z.



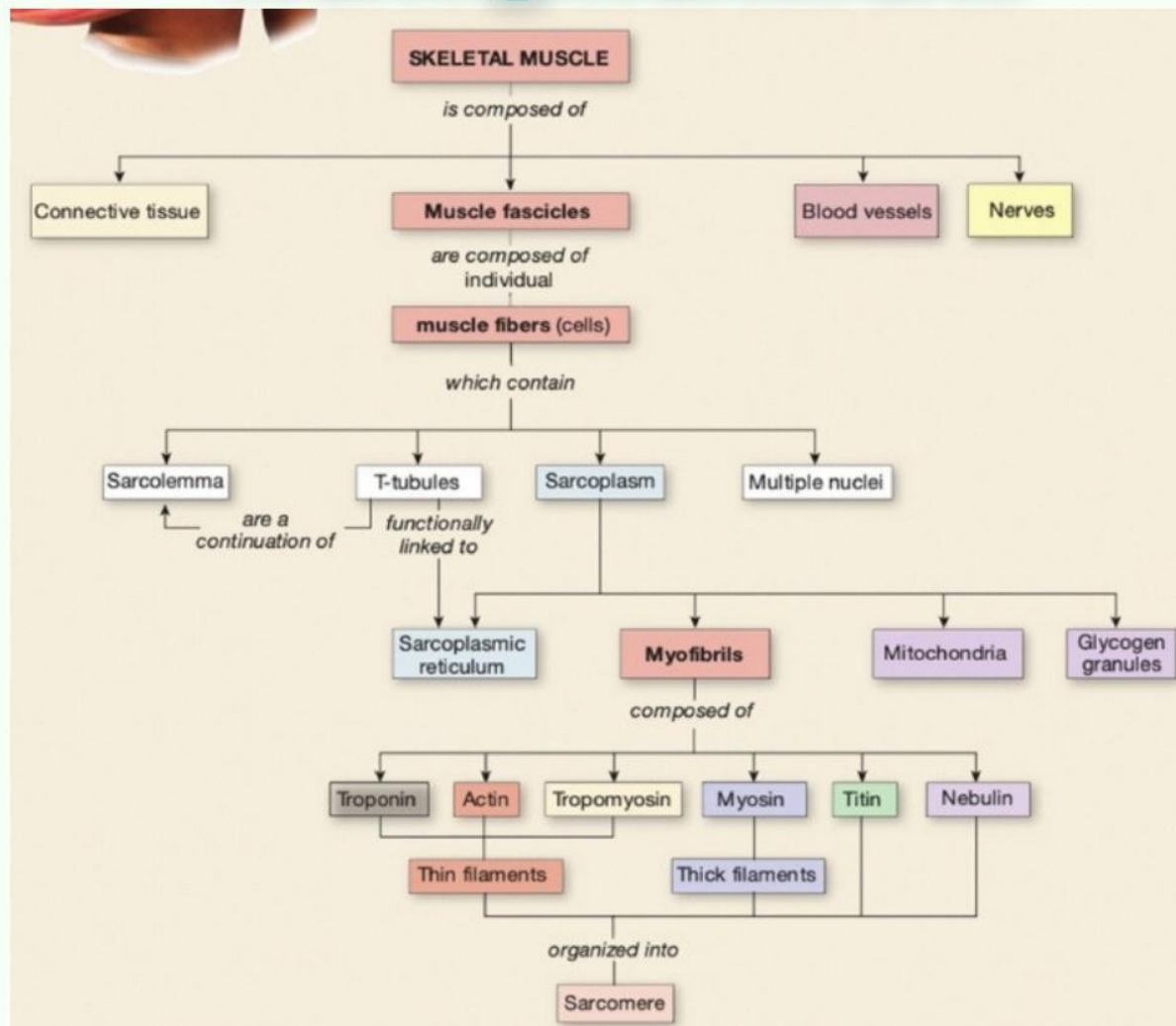
Terminologi

Komposisi Filamen

- Protein kontraktil: aktin dan miosin
- Protein regulatorik: troponin dan tropomiosin
- Protein struktural: titin, nebulin, myomesin, dystrophin



Rangkuman



Jenis-Jenis Otot



Jenis-Jenis Otot

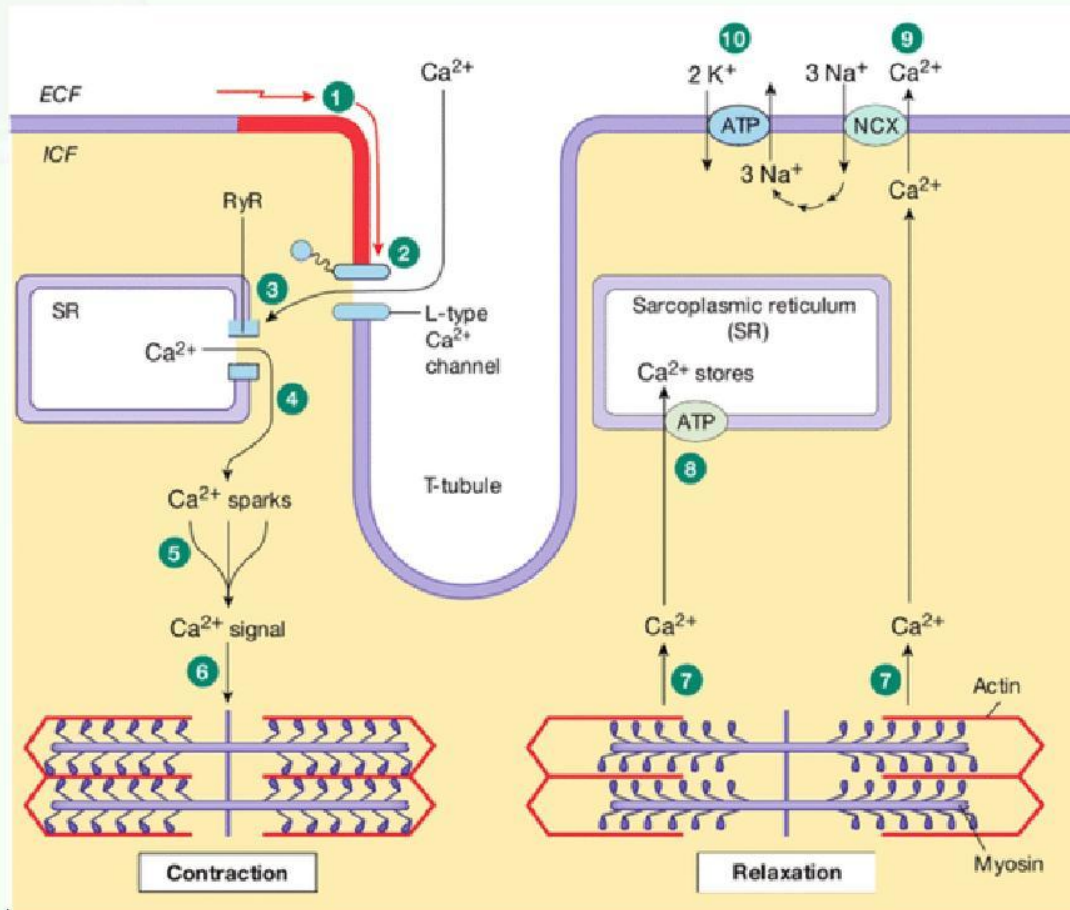
	SKELETAL	SMOOTH	CARDIAC
Appearance under light microscope	Striated	Smooth	Striated
Fiber arrangement	Sarcomeres	Oblique bundles	Sarcomeres
Location	Attached to bones; a few sphincters close off hollow organs	Forms the walls of hollow organs and tubes; some sphincters	Heart muscle
Tissue morphology	Multinucleate; large, cylindrical fibers	Uninucleate; small spindle-shaped fibers	Uninucleate; shorter branching fibers
Internal structure	T-tubule and sarcoplasmic reticulum	No t-tubules; sarcoplasmic reticulum reduced or absent	T-tubule and sarcoplasmic reticulum
Fiber proteins	Actin, myosin; troponin and tropomyosin	Actin, myosin, tropomyosin	Actin, myosin; troponin and tropomyosin
Control	• Ca^{2+} and troponin • Fibers independent of one another	• Ca^{2+} and calmodulin • Fibers electrically linked via gap junctions	• Ca^{2+} and troponin • Fibers electrically linked via gap junctions
Contraction speed	Fastest	Slowest	Intermediate
Contraction force of single fiber twitch	Not graded	Graded	Graded
Initiation of contraction	Requires ACh from motor neuron	Stretch, chemical signals. Can be autorhythmic	Autorhythmic
Neural control of contraction	Somatic motor neuron	Autonomic neurons	Autonomic neurons
Hormonal influence on contraction	None	Multiple hormones	Epinephrine

Mekanisme ✨ Kontraksi Otot



Mekanisme Kontraksi Otot

EXCITATION-CONTRACTION COUPLING



- 1 Action potential enters from adjacent cell.
- 2 Voltage-gated Ca^{2+} channels open. Ca^{2+} enters cell.
- 3 Ca^{2+} induces Ca^{2+} release through ryanodine receptor-channels (RyR).
- 4 Local release causes Ca^{2+} spark.
- 5 Summed Ca^{2+} sparks create a Ca^{2+} signal.
- 6 Ca^{2+} ions bind to troponin to initiate contraction.
- 7 Relaxation occurs when Ca^{2+} unbinds from troponin.
- 8 Ca^{2+} is pumped back into the sarcoplasmic reticulum for storage.
- 9 Ca^{2+} is exchanged with Na^{+} by the NCX antiporter.
- 10 Na^{+} gradient is maintained by the Na^{+} - K^{+} -ATPase.

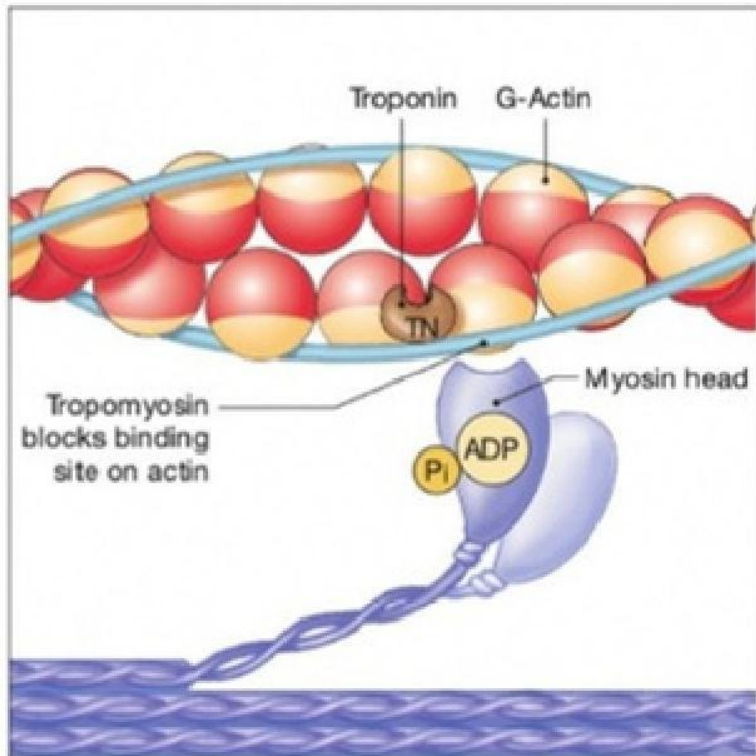
Kontraksi otot bersifat **All or None**/kontraksi otot hanya akan terjadi jika stimulus dapat mencapai ambang rangsang otot (threshold).

Proses singkat :

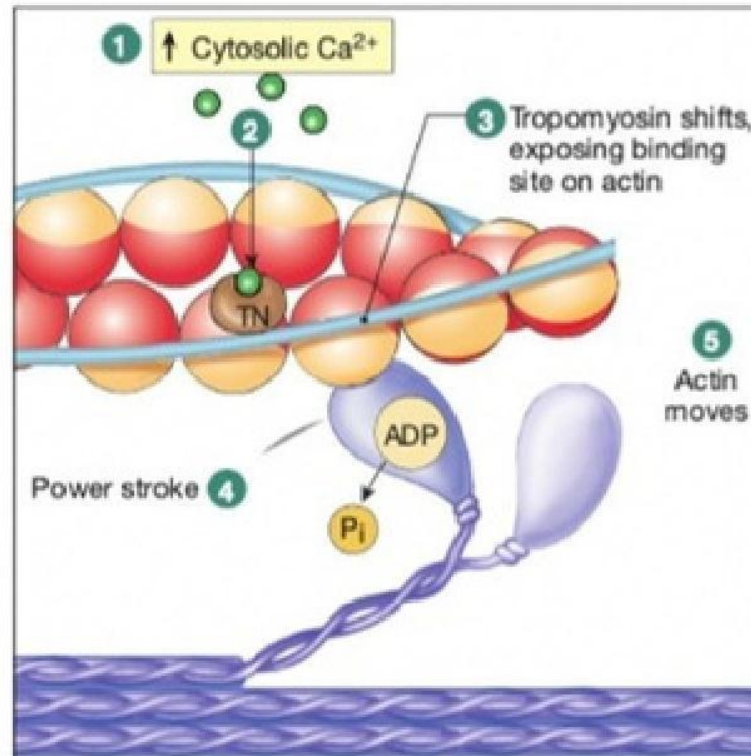
1. Pelepasan Acetylcholine (ACh) dari saraf motor somatik
2. ACh menimbulkan penyaluran Potensial Aksi
3. Potensial aksi membuka channel Ca^{2+}
4. Ca^{2+} menginisiasi kontraksi otot

Mekanisme Kontraksi Otot

Kalsium Menginisiasi Kontraksi Otot



(a) Relaxed state. Myosin head cocked.



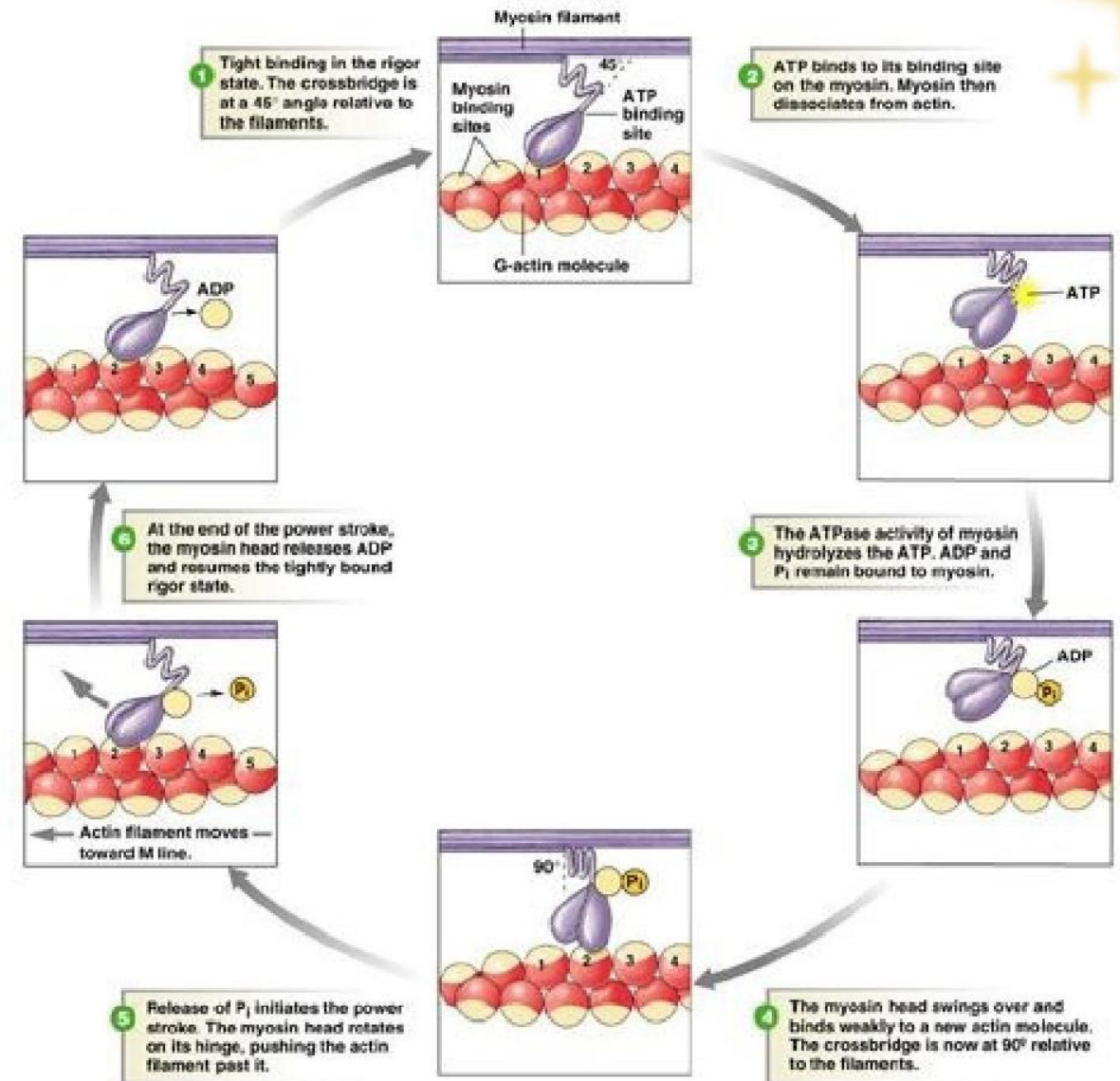
(b) Initiation of contraction

- 1 Ca^{2+} levels increase in cytosol.
- 2 Ca^{2+} binds to troponin (TN).
- 3 Troponin- Ca^{2+} complex pulls tropomyosin away from actin's myosin-binding site.
- 4 Myosin binds to actin and completes power stroke.
- 5 Actin filament moves.

Mekanisme Kontraksi Otot

Peran ATP

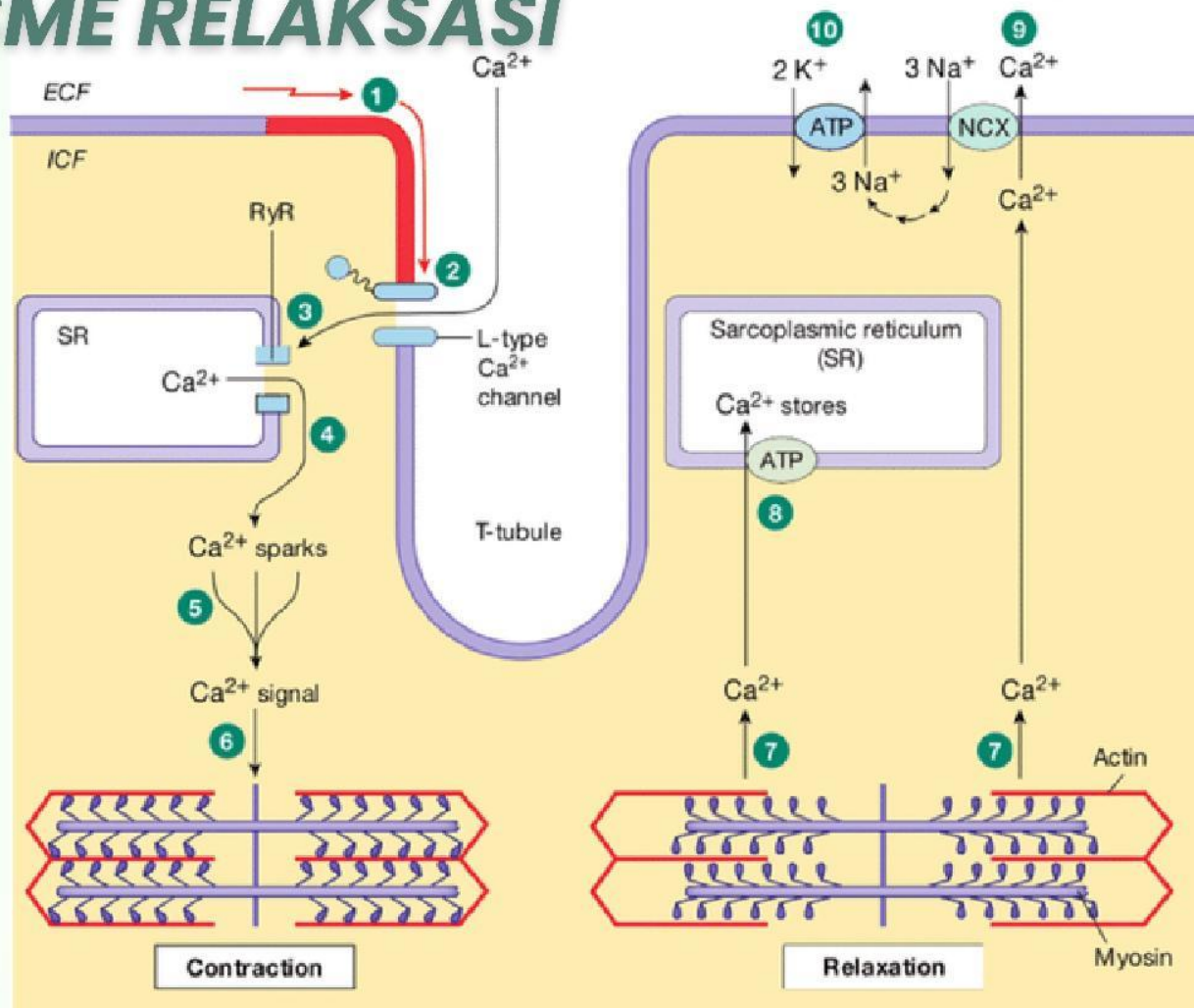
- Siklus diawali dari fase rigor/kaku
- ATP akan melepaskan ikatan aktin-miosin
- ATP dihidrolisis menjadi ADP dan P_i /fosfat
- Miosin kembali berikatan dengan aktin
- Saat ada sinyal Ca^{2+} , terjadi crossbridge tilting/power stroke bersamaan dengan pelepasan P_i
- ADP lepas dari kepala miosin
- Kembali ke fase rigor



Mekanisme Kontraksi Otot

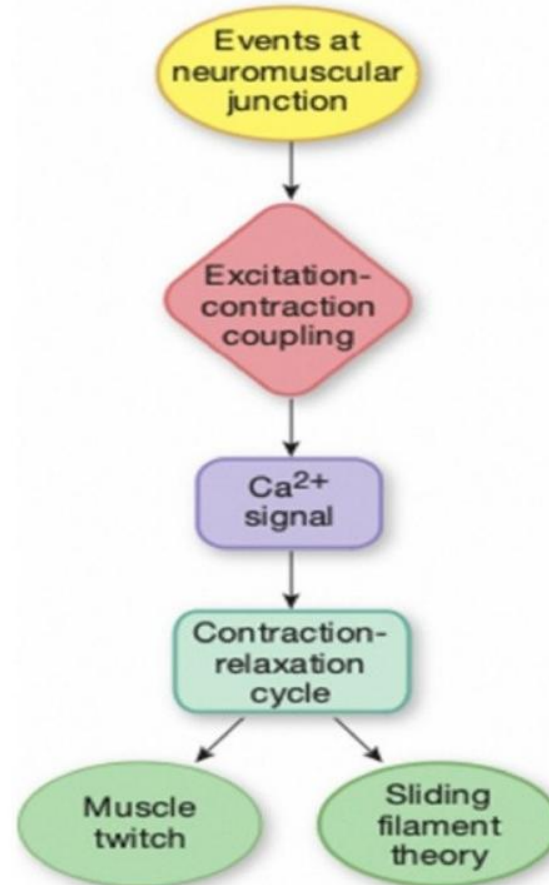
MEKANISME RELAKSASI

Step 7-10

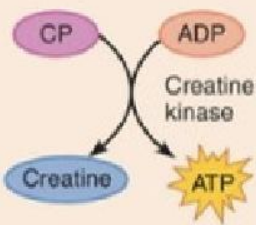
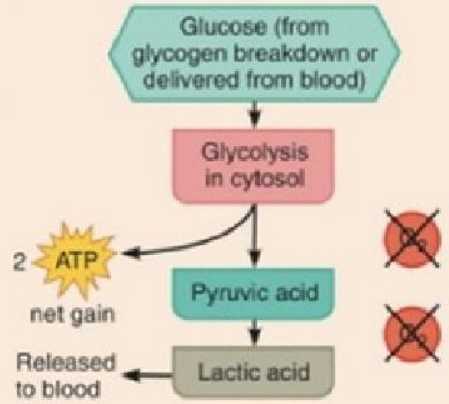
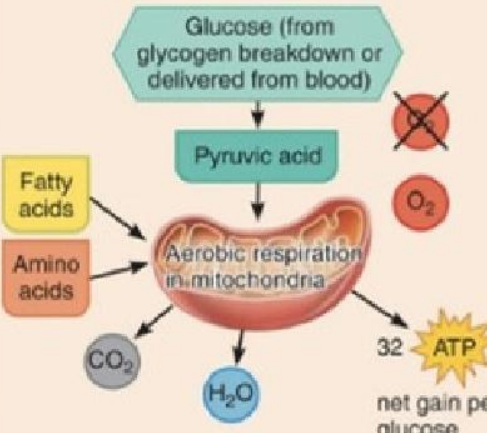


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Rangkuman



Sumber Energi Otot

(a) Direct phosphorylation	(b) Anaerobic pathway	(c) Aerobic pathway
Coupled reaction of creatine phosphate (CP) and ADP Energy source: CP  Oxygen use: None Products: 1 ATP per CP, creatine Duration of energy provided: 15 seconds	Glycolysis and lactic acid formation Energy source: glucose  Oxygen use: None Products: 2 ATP per glucose, lactic acid Duration of energy provided: 30–40 seconds, or slightly more	Aerobic cellular respiration Energy source: glucose; pyruvic acid; free fatty acids from adipose tissue; amino acids from protein catabolism  Oxygen use: Required Products: 32 ATP per glucose, CO₂, H₂O Duration of energy provided: Hours

A. Direct phosphorylation

- Olahraga berintensitas tinggi dengan beban berat (sprint, weight lifting)

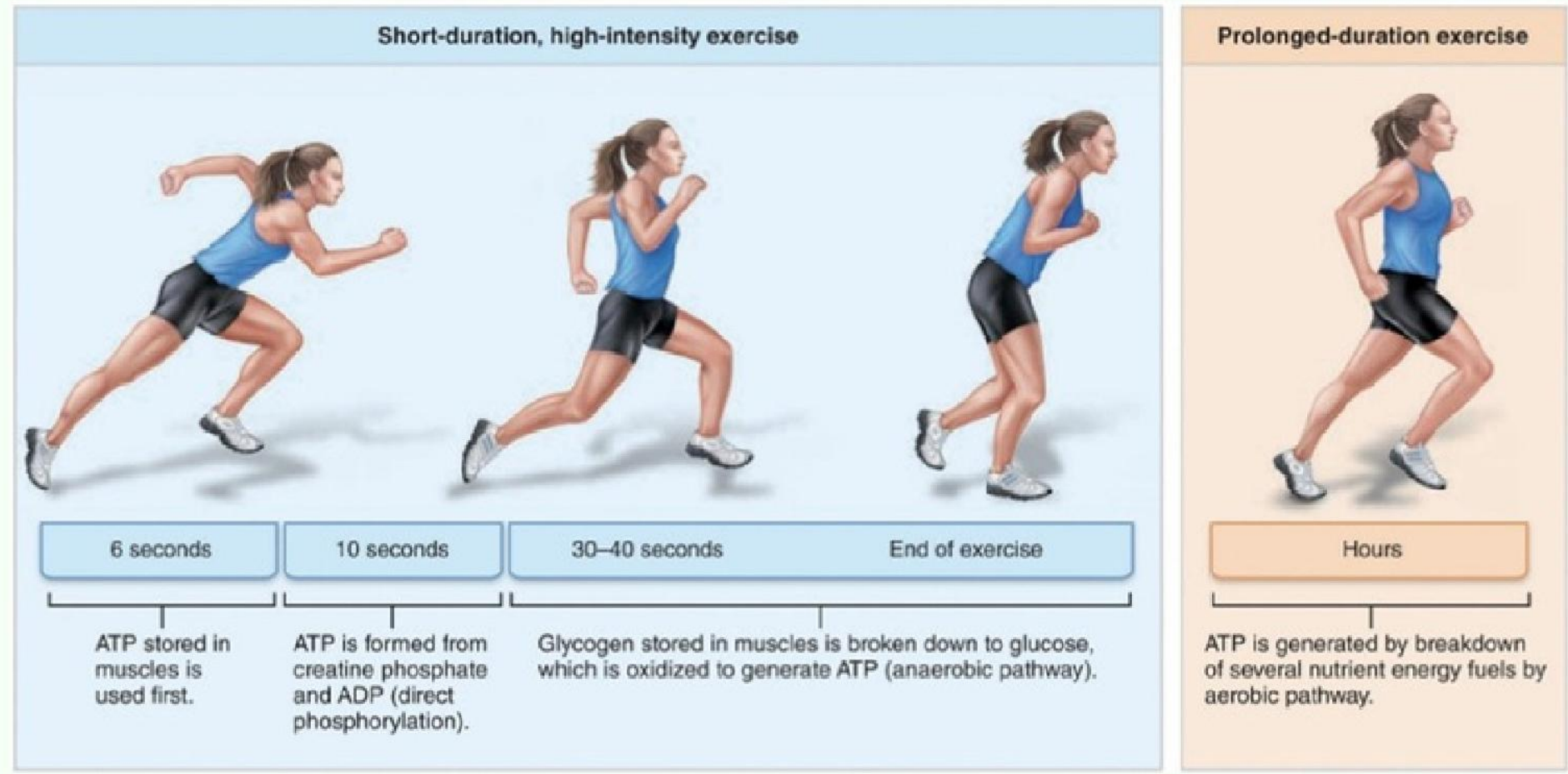
B. Anaerobic pathway

- Tenis, sepakbola, renang 100 meter

C. Aerobic pathway

- Lari maraton dan jogging

Sumber Energi Otot



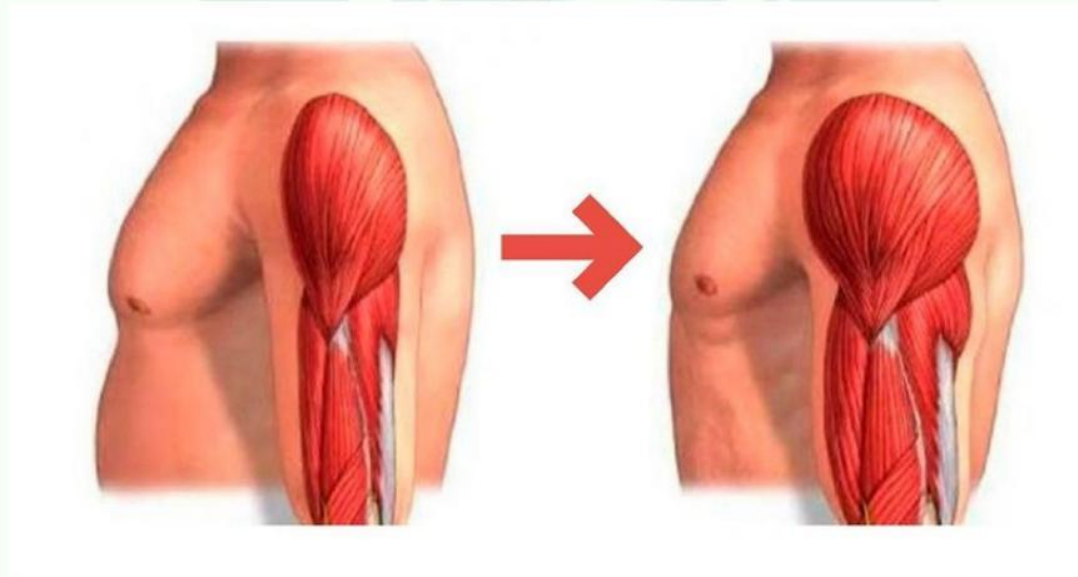
Korelasi Klinis



Korelasi Klinis

MUSCLE HYPERTROPHY

- Pertambahan ukuran otot karena penambahan jumlah myosin dan aktin.
- Dapat terjadi ketika melakukan olahraga ketahanan tingkat tinggi secara rutin, dalam waktu singkat sehingga hanya menggunakan mekanisme anaerob (glikolisis).
 - Ex: angkat beban

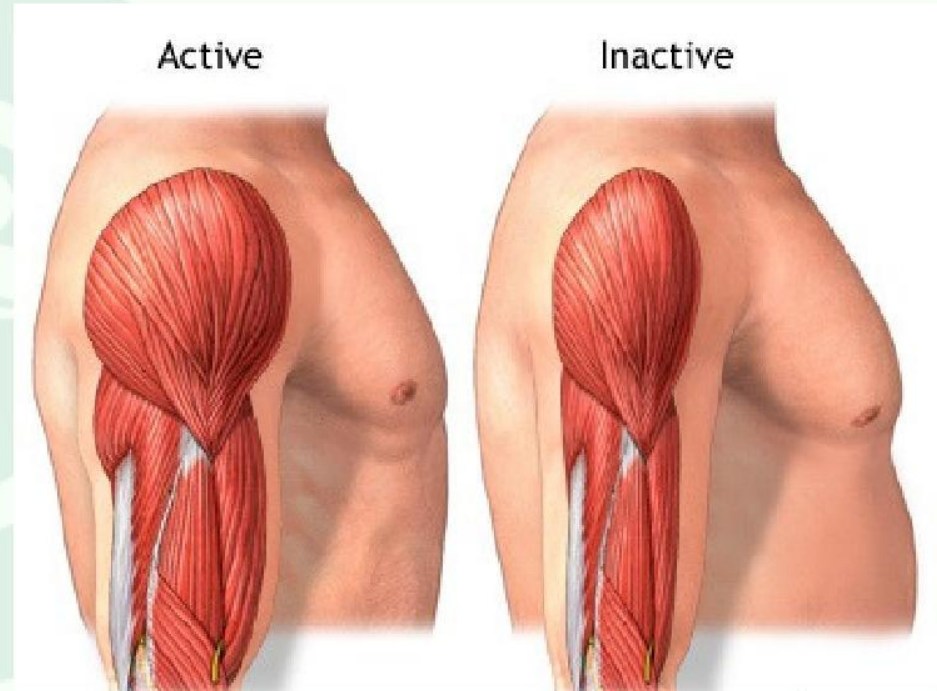


Korelasi Klinis

MUSCLE ATROPHY

→ Pengurangan aktin dan myosin karena lama tidak digunakan sehingga ukuran otot mengecil dan lemah.

- Dapat terjadi karena berbaring cukup lama (pasien di rumah sakit yang bertahun-tahun), saraf yang tidak intact/hilang sehingga otot tidak berkontraksi, ataupun faktor usia (sarkopenia).



Korelasi Klinis

RIGOR MORTIS

- Kekakuan otot pada mayat yang sudah meninggal
- Karena tidak adanya produksi ATP untuk melepas ikatan aktin dan myosin (crossbridge formation tidak bisa lepas)
 - Merupakan kondisi fisiologis



Korelasi Klinis

MUSCLE CRAMP/CHARLEY HORSE

→ Kontraksi otot secara terus-menerus, tidak sadar, dan terasa nyeri

- Terjadi karena ada hipereksitabilitas saraf somatik motorik yang mengontrol otot tersebut
- Bisa terjadi dalam waktu detik sampai menit
- Bisa diatasi dengan stretching
- Penyebab: kekurangan elektrolit, keringat berlebihan



Soal Bonus



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- Silverthorn, D. U. (2019). Human Physiology: An Integrated Approach (8th ed.). Pearson.





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